

7 Telephony and SMS

Let's see how to use Android's telephony APIs to monitor mobile voice and data connections as well as incoming and outgoing calls, and to send and receive SMS messages.

Android provides full access to SMS functionality, letting you send and receive SMS messages from within your applications. Using the Android APIs, you can create your own SMS client application to replace the native clients available as part of the software stack. Alternatively, you can incorporate the messaging functionality within your own applications to create social applications using SMS as the transport layer.

7.1 Handling Telephone Calls

Android has the means to let you manipulate the phone just like any other piece of the Android system.

7.2 Report to the Manager

To get at much of the phone API, you use the `TelephonyManager`. That class lets you do things like the following:

- Determine if the phone is in use via `getCallState()`, with return values of `CALL_STATE_IDLE` (phone not in use), `CALL_STATE_RINGING` (call requested but still being connected), and `CALL_STATE_OFFHOOK` (call in progress).
- Find out the SIM ID (IMSI) via `getSubscriberId()`.
- Find out the phone type (e.g., GSM) via `getPhoneType()`, or find out the data connection type (e.g., GPRS or EDGE) via `getNetworkType()`.

You can also initiate a call from your application, such as from a phone number you obtained through your own web service. To do this, simply craft an `ACTION_DIAL` Intent with a Uri of the form `tel:NNNNN` (where NNNNN is the phone number to dial) and use that Intent with `startActivity()`. This will not actually dial the phone; rather, it activates the dialer activity, from which the user can press a button to place the call.

For example, let's look at the Phone/Dialer sample application.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/
android"
android:orientation="vertical"
```